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AGR 101: Principles of Agriculture - Soils, Crops and Farming Systems

The science behind food production, from soil physics to farm planning

FACULTY PRIMER

Faculty Primer

Faculty of Agriculture
Crop Science & Horticulture | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Independent study support | Not an official UNIZIK publication

START HERE

Why this pack exists

This pack introduces agriculture as applied science and business: soil formation and fertility, crop growth and management, animal production basics, farming systems and the realities of extension and markets in Nigeria.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Describe soil formation, texture, structure and fertility management.
- Explain the growth stages and management needs of a field crop.
- Compare farming systems and their suitability to Nigerian conditions.
- Identify the main constraints in Nigerian agricultural value chains.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

Soil as a living system

Soil is the medium of plant growth, and it is a biological system, not inert dirt.

- Soil formation involves weathering, organic matter accumulation and biological activity over time.
- Texture refers to the proportions of sand, silt and clay; structure refers to how particles aggregate.
- Porosity and water-holding capacity determine irrigation and drainage needs.
- Soil pH affects nutrient availability; most crops do best in a slightly acidic to neutral range.
- Organic matter improves structure, water retention and microbial activity.
- Fertility requires both nutrients and suitable physical and biological conditions.

> **Test before you apply**
> Applying fertiliser without a soil test is guesswork. Excess nitrogen can burn crops and pollute water, while a micronutrient deficiency may go untreated.

Crop production

Crop yield is the product of genetics, environment and management.

- Land preparation: clearing, tillage and seedbed formation appropriate to the crop.
- Propagation: seed selection, nursery practice, transplanting and spacing.
- Water management: irrigation scheduling based on crop need and soil moisture.
- Weed control: mechanical, cultural, biological and chemical methods, and integrated management.
- Pest and disease management: scouting, thresholds, resistant varieties and safe pesticide use.
- Post-harvest handling determines whether the yield reaches the market in saleable condition.

Systems, livestock and value chains

A farm is a system in which crops, animals, labour and money interact.

- Shifting cultivation and bush fallow rely on long fallow periods to restore fertility.
- Mixed cropping and intercropping spread risk and use space efficiently.
- Intensive systems require higher inputs and careful management.
- Livestock production depends on breeds adapted to local conditions, feeding, housing and health.
- Extension links research to farmers but is constrained by staffing and funding.
- Value chain constraints include storage losses, poor roads, credit scarcity and price volatility.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. What is soil texture?

Answer: The relative proportions of sand, silt and clay particles in a soil.

2. Why does organic matter matter?

Answer: It improves soil structure, water-holding capacity and biological activity, and supplies nutrients as it decomposes.

3. Give one advantage and one disadvantage of mixed cropping.

Answer: Advantage: risk spreading and efficient land use. Disadvantage: competition for light, water and nutrients and greater complexity in pest management.

4. What is integrated pest management?

Answer: A strategy combining cultural, biological, mechanical and chemical methods with monitoring and thresholds, using chemicals as a last resort.

5. Name two post-harvest constraints in Nigeria.

Answer: Inadequate storage leading to spoilage, and poor road infrastructure leading to delayed and damaged transport.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Describe the four components of soil.
- [ ] Test the texture of two soil samples by hand feel and record results.
- [ ] List the growth stages of a common Nigerian crop.
- [ ] Compare three farming systems in a table.
- [ ] Identify five pests or diseases of a staple crop.
- [ ] Draw a simple value chain for a crop you know.
- [ ] Interview a farmer about constraints and record five findings.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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